

Laboratorio #6.2: Escalado en Kubernetes con Flask (Deployment, Service y réplicas)

Objetivos:

- Desplegar una aplicación Flask en Kubernetes con un Deployment y exponerla con un Service NodePort.
- Escalar réplicas de forma imperativa (kubectl scale) y declarativa (YAML + kubectl apply).
- Observar cómo el Service balancea el tráfico entre las réplicas.
- Comprobar la autorreparación (self-healing) del ReplicaSet al eliminar un Pod.
- Configurar autoescalado con un HorizontalPodAutoscaler.

Guías de laboratorio:

- https://github.com/wils0n/laboratorios-md/blob/main/laboratorio_kubernetes/laboratorio_kubernetes.md

Entregables

1. **URL de repositorio** (GitHub o GitLab).
 - o <https://github.com/aarontimana04/Lab06-2.git>
2. Subir su repositorio usando buenas prácticas de git (GitFlow/TBD, conventional commits, PR, aprobación con un revisor)

Estructura del repo:

Ejemplo:

```
flask-k8s-app/
├── app/
│   ├── app.py
│   ├── requirements.txt
│   └── Dockerfile
├── k8s/
│   ├── 01-deployment.yaml
│   └── 02-service.yaml
├── scripts/
│   └── balanceo.sh
└── README.md
```

3. Captura de kubectl get pods -o wide con los 2 Pods de Flask en Running y READY 1/1 (parte 2.1).

```

usuario@MacBookPro Lab06-2 % kubectl get deployment
NAME      READY   UP-TO-DATE   AVAILABLE   AGE
flask-app 0/2     2            0           2m14s
usuario@MacBookPro Lab06-2 % kubectl get pods -o wide
NAME                                READY   STATUS    RESTARTS   AGE   IP            NODE           NOMINATED NODE   READINESS GATES
flask-app-d894dc69d-4596w          0/1     CrashLoopBackOff   4 (3s ago)  2m34s  10.1.0.7      docker-desktop  <none>            <none>
flask-app-d894dc69d-pg82p          0/1     CrashLoopBackOff   4 (3s ago)  2m34s  10.1.0.6      docker-desktop  <none>            <none>
usuario@MacBookPro Lab06-2 %

```

4. Captura de ./scripts/balanceo.sh 10 mostrando el conteo final "Peticiones por Pod" con 2 Pods distintos (parte 2.3).

```

usuario@MacBookPro Lab06-2 % ./scripts/balanceo.sh 10
== 10 peticiones a http://localhost:30080/pod ==

peticion 1 -> flask-app-688c977c4b-99qdl
peticion 2 -> flask-app-688c977c4b-99qdl
peticion 3 -> flask-app-688c977c4b-99qdl
peticion 4 -> flask-app-688c977c4b-99qdl
peticion 5 -> flask-app-688c977c4b-99qdl
peticion 6 -> flask-app-688c977c4b-99qdl
peticion 7 -> flask-app-688c977c4b-99qdl
peticion 8 -> flask-app-688c977c4b-99qdl
peticion 9 -> flask-app-688c977c4b-99qdl
peticion 10 -> flask-app-688c977c4b-99qdl

== Peticiones por Pod ==
10 flask-app-688c977c4b-99qdl
usuario@MacBookPro Lab06-2 %

```

5. Captura de kubectl get deployment flask-app con READY 5/5 y del comando kubectl get endpointslices (jsonpath de la guía) listando las 5 IPs tras kubectl scale (parte 3.1).

```

usuario@MacBookPro Lab06-2 % kubectl get deployment flask-app
NAME      READY   UP-TO-DATE   AVAILABLE   AGE
flask-app 5/5     5            5           41m
usuario@MacBookPro Lab06-2 % kubectl get deployment flask-app
NAME      READY   UP-TO-DATE   AVAILABLE   AGE
flask-app 5/5     5            5           41m
usuario@MacBookPro Lab06-2 % kubectl get endpointslices \
-l kubernetes.io/service-name=flask-service \
-o jsonpath='{range .items[*].endpoints[*]}.{.addresses[0]}{"\n"}{end}'
10.1.0.8
10.1.0.9
10.1.0.11
10.1.0.12
10.1.0.10
usuario@MacBookPro Lab06-2 %

```

6. Captura de kubectl get pods justo después de eliminar un Pod, mostrando el Pod recreado con edad de pocos segundos (parte 3.3).

```

usuario@MacBookPro Lab06-2 % nano k8s/01-deployment.yaml
usuario@MacBookPro Lab06-2 % kubectl apply -f k8s/01-deployment.yaml
deployment.apps/flask-app configured
usuario@MacBookPro Lab06-2 % kubectl get deployment flask-app
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
flask-app     3/3     3             3           43m
usuario@MacBookPro Lab06-2 % kubectl get deployment flask-app
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
flask-app     3/3     3             3           44m
usuario@MacBookPro Lab06-2 % kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
flask-app-688c977c4b-99qd1  1/1     Running   0           6m21s
flask-app-688c977c4b-hj5l1  1/1     Running   0           3m41s
flask-app-688c977c4b-jlgr5  1/1     Running   1 (4m33s ago) 6m25s
usuario@MacBookPro Lab06-2 % POD=$(kubectl get pods -l app=flask -o jsonpath='{.items[0].metadata.name}')
usuario@MacBookPro Lab06-2 % echo "Eliminando $POD"
Eliminando flask-app-688c977c4b-99qd1
usuario@MacBookPro Lab06-2 % kubectl delete pod $POD
pod "flask-app-688c977c4b-99qd1" deleted
usuario@MacBookPro Lab06-2 % kubectl get pods
NAME          READY   STATUS    RESTARTS   AGE
flask-app-688c977c4b-hj5l1  1/1     Running   0           4m37s
flask-app-688c977c4b-jlgr5  1/1     Running   1 (5m29s ago) 7m21s
flask-app-688c977c4b-x5jd1  0/1     Running   0           7s
usuario@MacBookPro Lab06-2 %

```

7. (Reto, parte 5) Captura de kubectl get hpa mostrando la columna REPLICAS subiendo bajo carga.

```

Lab06-2 ~ curl -s http://localhost:30080 -- 126x70
[!] failed to verify certificate x809: cannot validate certificate for 192.168.65.3 because it doesn't contain any IP SANs' no
de=docker-desktop
1 server:go:192] 'failed probe' probe='metric-storage-ready' err='no metrics to serve'
1 server:go:160] 'failed to scrape node' err='Get "https://192.168.65.3:10250/metrics/resource?":
1 server:go:160] 'failed to scrape node' err='Get "https://192.168.65.3:10250/metrics/resource?":
[!] failed to verify certificate x809: cannot validate certificate for 192.168.65.3 because it doesn't contain any IP SANs' no
de=docker-desktop
1 server:go:192] 'failed probe' probe='metric-storage-ready' err='no metrics to serve'
1 server:go:160] 'failed to scrape node' err='Get "https://192.168.65.3:10250/metrics/resource?":
1 server:go:160] 'failed to scrape node' err='Get "https://192.168.65.3:10250/metrics/resource?":
[!] failed to verify certificate x809: cannot validate certificate for 192.168.65.3 because it doesn't contain any IP SANs' no
de=docker-desktop
usuario@MacBookPro Lab06-2 % kubectl edit deployment metrics-server -n kube-system
error: there was a problem with the editor: vim
usuario@MacBookPro Lab06-2 % kubectl rollout restart deployment metrics-server -n kube-system
deployment.apps/metrics-server restarted
usuario@MacBookPro Lab06-2 % kubectl get pods -n kube-system | grep metrics
metrics-server-667c75d8-55dg  0/1     Running   0           18s
metrics-server-7b7c6b4d-vgmf  0/1     Running   0           30ms
usuario@MacBookPro Lab06-2 % kubectl patch deployment metrics-server -n kube-system --type=json -p '[
  {
    "op": "add",
    "path": "/spec/template/spec/containers/0/args/-",
    "value": "--kubelet-insecure-tls"
  }
]'
deployment.apps/metrics-server patched
usuario@MacBookPro Lab06-2 % kubectl rollout restart deployment metrics-server -n kube-system
deployment.apps/metrics-server restarted
usuario@MacBookPro Lab06-2 % kubectl get pods -n kube-system -w
NAME          READY   STATUS    RESTARTS   AGE
coradns-668d6f9bc-d9ap  1/1     Running   0           108m
coradns-668d6f9bc-p2qp  1/1     Running   0           108m
etcd-docker-desktop  1/1     Running   0           108m
kube-apiserver-docker-desktop  1/1     Running   0           108m
kube-controller-manager-docker-desktop  1/1     Running   0           108m
kube-proxy-dontop  1/1     Running   0           108m
kube-scheduler-docker-desktop  1/1     Running   0           108m
metrics-server-6d6b9746c-mdbq  0/1     Running   0           5s
metrics-server-784dc4d8f-prpm  0/1     Running   0           7s
storage-provisioner  1/1     Running   0           108m
vsnet-controller  1/1     Running   0           108m
metrics-server-6d6b9746c-mdbq  1/1     Running   0           22s
metrics-server-784dc4d8f-prpm  1/1     Running   0           22s
metrics-server-6d6b9746c-mdbq  1/1     Terminating 0           30s
metrics-server-6d6b9746c-mdbq  0/1     Completed 0           32s
metrics-server-6d6b9746c-mdbq  0/1     Completed 0           32s
metrics-server-6d6b9746c-mdbq  0/1     Completed 0           32s
usuario@MacBookPro Lab06-2 % kubectl top nodes
NAME          CPU(cores)   CPU(N)      MEMORY(bytes)  MEMORY(N)
docker-desktop  45m          0N          1320M          17%
usuario@MacBookPro Lab06-2 % kubectl top pods
NAME          CPU(cores)   MEMORY(bytes)
flask-app-688c977c4b-hj5l1  1m           32M
flask-app-688c977c4b-jlgr5  32m          32M
flask-app-688c977c4b-x5jd1  1m           32M
usuario@MacBookPro Lab06-2 % kubectl get hpa
No resources found in default namespace.
usuario@MacBookPro Lab06-2 % kubectl autoscale deployment flask-app \
--cpu-percent=20 \
--min=1 \
--max=18
horizontalpodautoscaler.autoscaling/flask-app autoscaled
usuario@MacBookPro Lab06-2 % kubectl get hpa
NAME          REFERENCE   TARGETS   MINPODS   MAXPODS   REPLICAS   AGE
flask-app     Deployment/flask-app  cpu: unknown/20%  2          18          0           8s
usuario@MacBookPro Lab06-2 % while true; do
curl -s http://localhost:30080/ > /dev/null
done

```